

Incremental Loading in Azure Data Factory

Introduction

This project demonstrates an **Incremental Data Loading** process using **Azure Data Factory, Azure SQL Database, and Azure Data Lake Storage Gen2**.

The project loads only the **new or modified records** from the source instead of loading the complete data every time. A **watermark value** is used to identify the records that need to be loaded.

Tools & Services Used

- Azure Data Factory
- Azure SQL Database
- Azure Data Lake Storage Gen2
- SQL
- Stored Procedure
- Lookup Activity
- Copy Data Activity

Project Flow

Azure SQL Database → Azure Data Factory → ADLS Gen2

The pipeline uses Lookup activities to identify the previous and latest watermark values, Copy Data activity to transfer the incremental records, and a Stored Procedure to update the watermark value.

Key Steps

1. Create and configure Azure resources.
2. Create the source table and watermark table in SQL Database.
3. Configure Linked Services and Datasets in ADF.
4. Use Lookup activities to retrieve watermark values.
5. Use Copy Data activity to load incremental records.
6. Use Stored Procedure activity to update the watermark.
7. Validate and debug the pipeline.
8. Verify the successful data transfer.

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Open Azure Portal and open the required Resource Group.

Verify the Azure resources used for the project.

The screenshot shows the Microsoft Azure Portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and user information for 'Nandana.VA@akumenb...'. The main content area displays the 'resource912' resource group. On the left, a sidebar lists various management tools like 'Overview', 'Activity log', 'Access control (IAM)', 'Tags', 'Resource visualizer', 'Events', 'Settings', 'Cost Management', 'Monitoring', 'Automation', 'Export template', and 'Help'. The main pane shows a list of resources under the 'Essentials' tab. The resources listed are:

Name	Type	Location
admin9	SQL server	Southeast Asia
factnanda	Data factory (V2)	East US
nanda76	Storage account	East US
sqldf (admin9/sqldf)	SQL database	Southeast Asia

At the bottom, it indicates 'Showing 1 - 4 of 4. Display count: auto' and a 'Give feedback' link.

Azure SQL Database → Query Editor.

Connect to the database to run SQL queries.

The screenshot shows the 'Query editor (preview)' interface for the 'sqldf (admin9/sqldf)' SQL database. The top navigation bar is consistent with the previous screenshot. The main content area features a 'New query' button and a 'New query' link. Below this, a message reads: 'Start exploring your data. Write and run SQL queries, create database objects with a template, or open a table or view to preview your data.' The left sidebar lists various management tools, including 'Query editor (preview)'. The Explorer pane on the right shows the database structure with 'sqldf' and 'dbo' tables.

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Create the required SQL table and insert the sample data.

Verify the records in the table.

The screenshot shows the Microsoft Azure portal interface for a SQL database. The left sidebar contains navigation options like Overview, Activity log, Tags, and Query editor (preview). The main area displays the 'SQL query 1' editor with the following SQL code:

```
1 create table person
2 (
3     PersonID int,
4     Name Varchar(200),
5     LastModifytime datetime
6 );
7
8 insert into person
9 (PersonID, Name, LastModifytime)
10 values
11 (1, 'Tiya', '7/1/2019 12:55:00 AM');
```

Below the query editor, the 'Results' tab shows the output of the query. It displays a table with three columns: PersonID, Name, and LastModifytime. The results are as follows:

PersonID	Name	LastModifytime
1	Tiya	2019-07-01T00:55:00.0000000
2	Miya	2019-07-02T05:24:00.0000000
3	Ziya	2019-07-03T02:36:00.0000000

The status bar at the bottom indicates 'Succeeded (0 sec 317 ms)' and 'Columns: 3 Rows: 5'.

Open the **Person** table and verify the stored records.

Check the LastModifyTime column.

The screenshot shows the Microsoft Azure portal interface for a SQL database. The left sidebar contains navigation options like Overview, Activity log, Tags, and Query editor (preview). The main area displays the 'SQL query 1' editor with the following SQL code:

```
7
8 insert into person
9 (PersonID, Name, LastModifytime)
10 values
11 (1, 'Tiya', '7/1/2019 12:55:00 AM'),
12 (2, 'Miya', '7/2/2019 5:24:00 AM'),
13 (3, 'Ziya', '7/3/2019 2:36:00 AM'),
14 (4, 'Miya', '7/4/2019 3:20:00 AM'),
15 (5, 'Ziya', '7/5/2019 8:05:00 AM');
16
17 select * from person;
```

Below the query editor, the 'Results' tab shows the output of the query. It displays a table with three columns: PersonID, Name, and LastModifytime. The results are as follows:

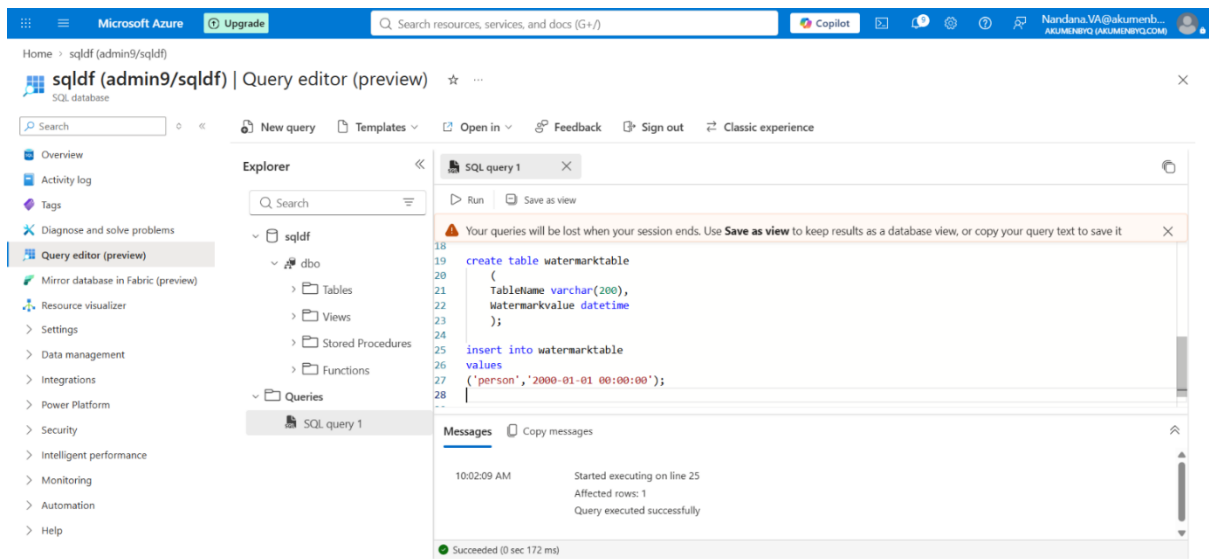
PersonID	Name	LastModifytime
1	Tiya	2019-07-01T00:55:00.0000000
2	Miya	2019-07-02T05:24:00.0000000
3	Ziya	2019-07-03T02:36:00.0000000
4	Miya	2019-07-02T05:24:00.0000000
5	Ziya	2019-07-02T05:24:00.0000000

The status bar at the bottom indicates 'Succeeded (0 sec 317 ms)' and 'Columns: 3 Rows: 5'.

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Create the **Watermark table**.

Insert the initial watermark value.



The screenshot shows the Microsoft Azure portal interface for the 'sqlrdf (admin9/sqlrdf)' database. The 'Query editor (preview)' is open, displaying a SQL script. The script creates a table named 'watermarktable' with columns 'TableName' (varchar(200)) and 'Watermarkvalue' (datetime). It then inserts a single row with the values 'person' and '2000-01-01 00:00:00'. The 'Messages' pane at the bottom indicates that the query executed successfully, affecting 1 row.

```
18
19
20 create table watermarktable
21 (
22     TableName varchar(200),
23     Watermarkvalue datetime
24 );
25
26 insert into watermarktable
27 values
28 ('person', '2000-01-01 00:00:00');
```

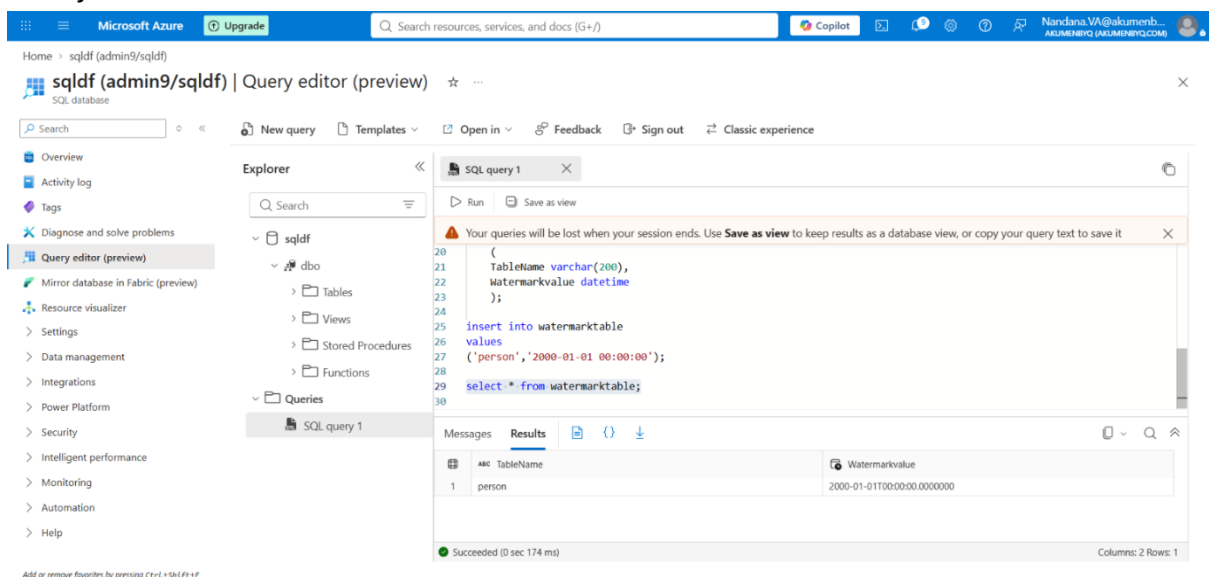
Messages

Time	Message
10:02:09 AM	Started executing on line 25 Affected rows: 1 Query executed successfully

Succeeded (0 sec: 172 ms)

Run a SELECT query on the watermark table.

Verify the stored TableName and WatermarkValue



The screenshot shows the Microsoft Azure portal interface for the 'sqlrdf (admin9/sqlrdf)' database. The 'Query editor (preview)' is open, displaying a SQL script. The script selects all data from the 'watermarktable'. The 'Results' pane at the bottom shows the output of the query, displaying a single row with the values 'person' and '2000-01-01T00:00:00.0000000'.

```
20
21 (
22     TableName varchar(200),
23     Watermarkvalue datetime
24 );
25
26 insert into watermarktable
27 values
28 ('person', '2000-01-01 00:00:00');
```

select * from watermarktable;

Results

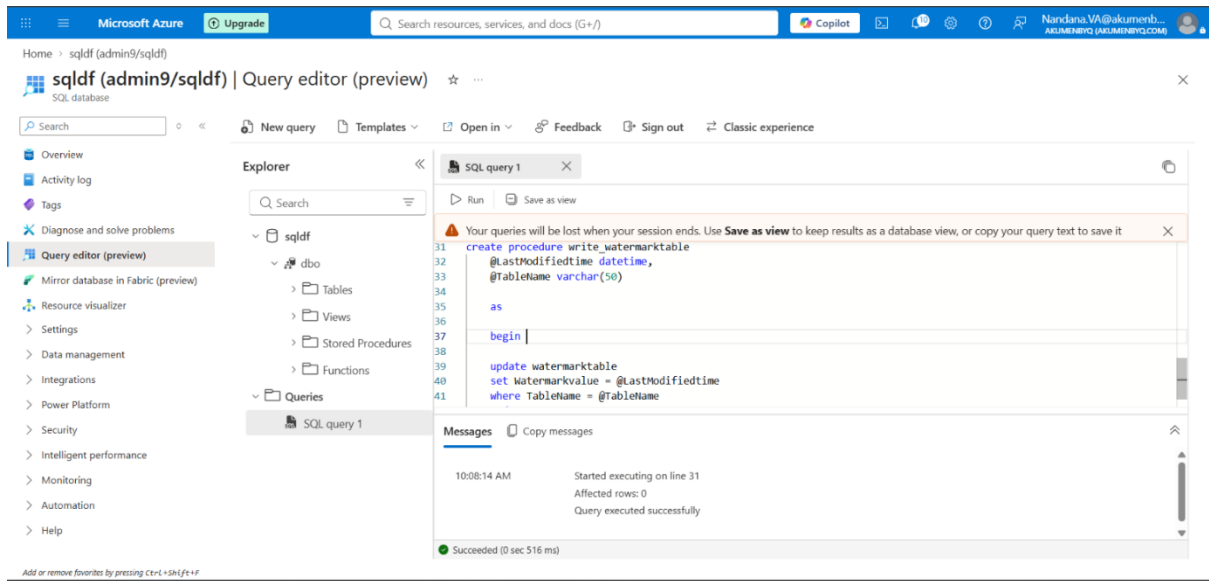
TableName	Watermarkvalue
person	2000-01-01T00:00:00.0000000

Succeeded (0 sec: 174 ms) Columns: 2 Rows: 1

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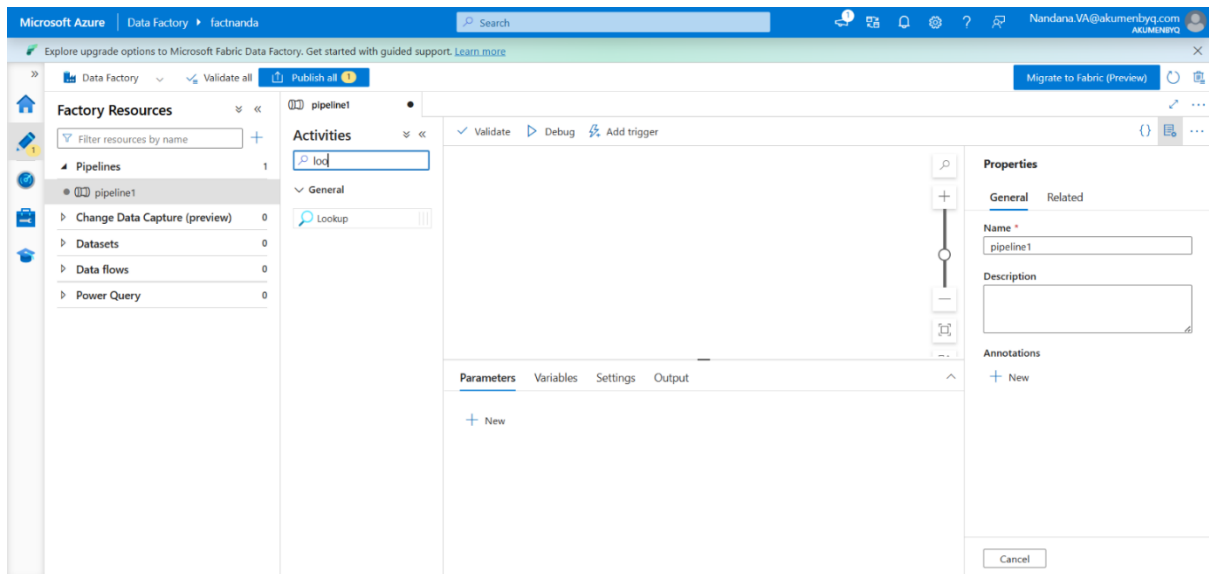
Create the **Stored Procedure** to update the watermark value.

Use LastModifyTime and TableName as required.



Open **Azure Data Factory**.

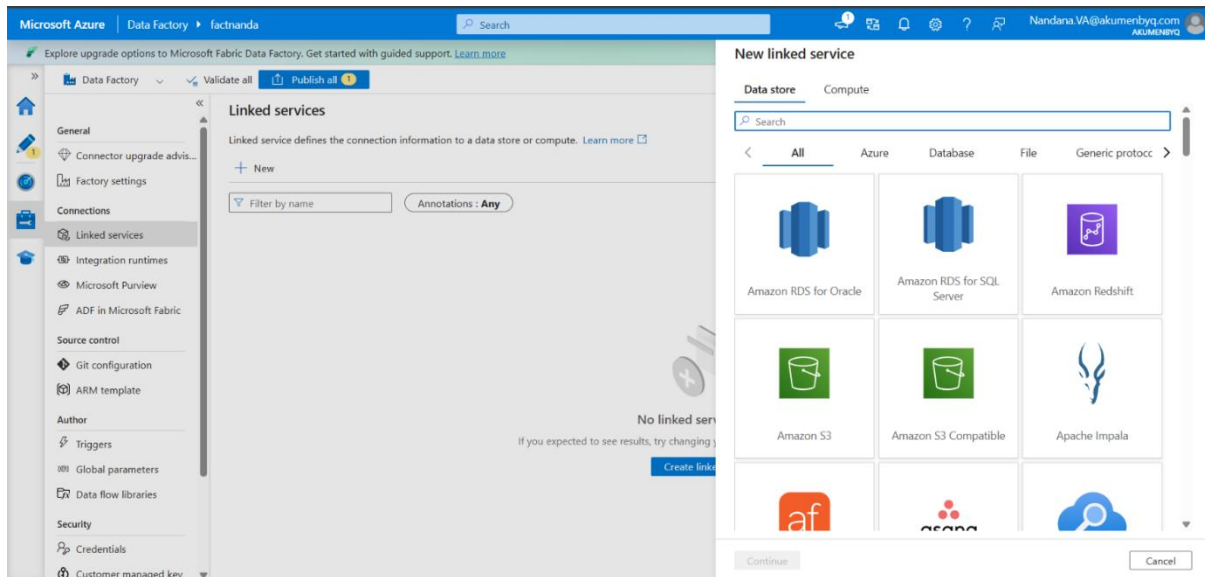
Create/open the Incremental Load pipeline and add a **Lookup activity**.



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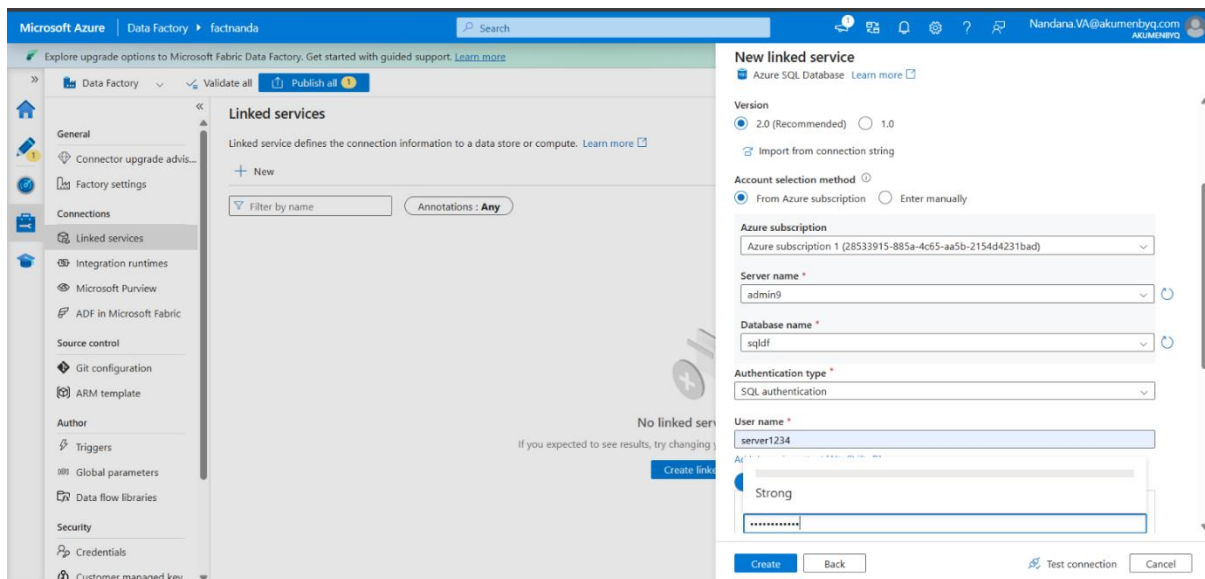
Click **New Linked Service** in ADF.

Select the required connector.



Create the **Azure SQL Database Linked Service**.

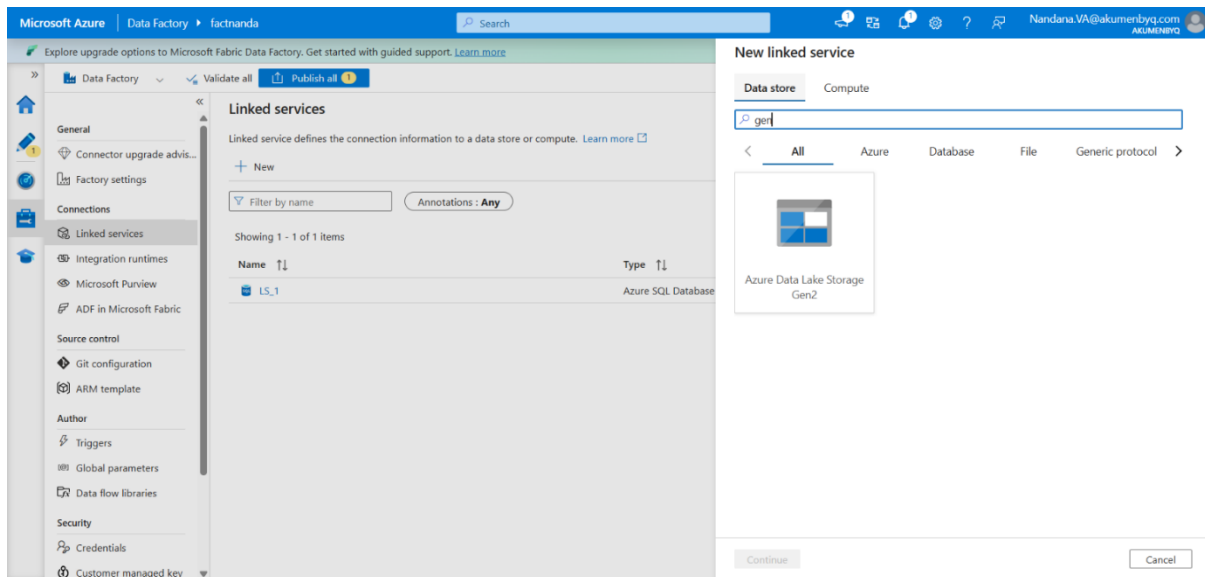
Enter the required database connection details.



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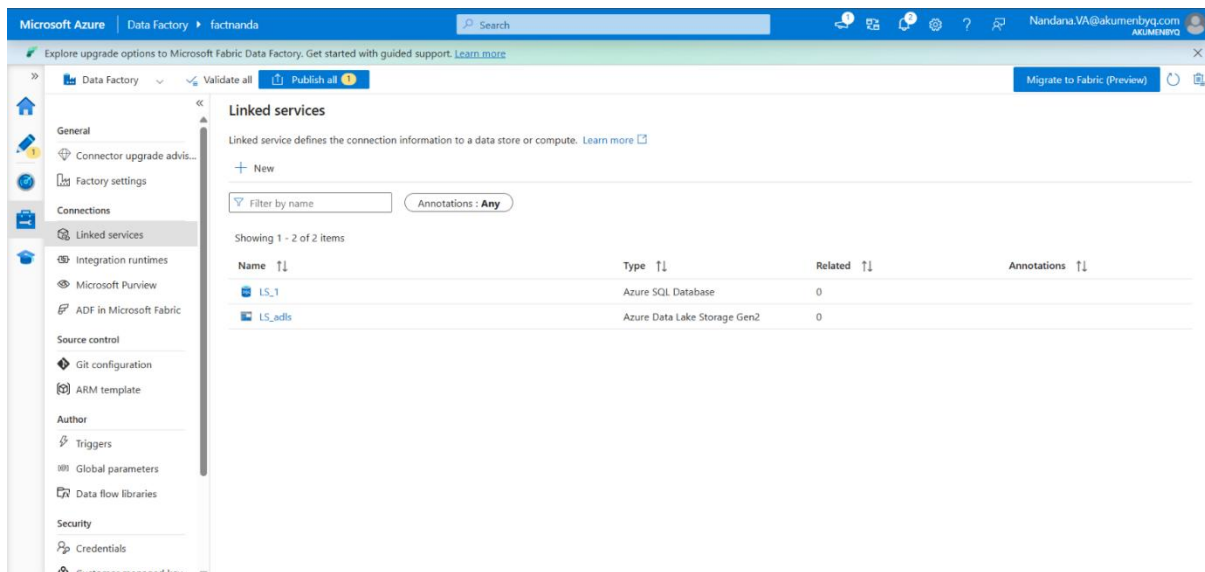
Select **Azure Data Lake Storage Gen2** as the connector.

Configure the ADLS connection.



Open **Manage** → **Linked Services**.

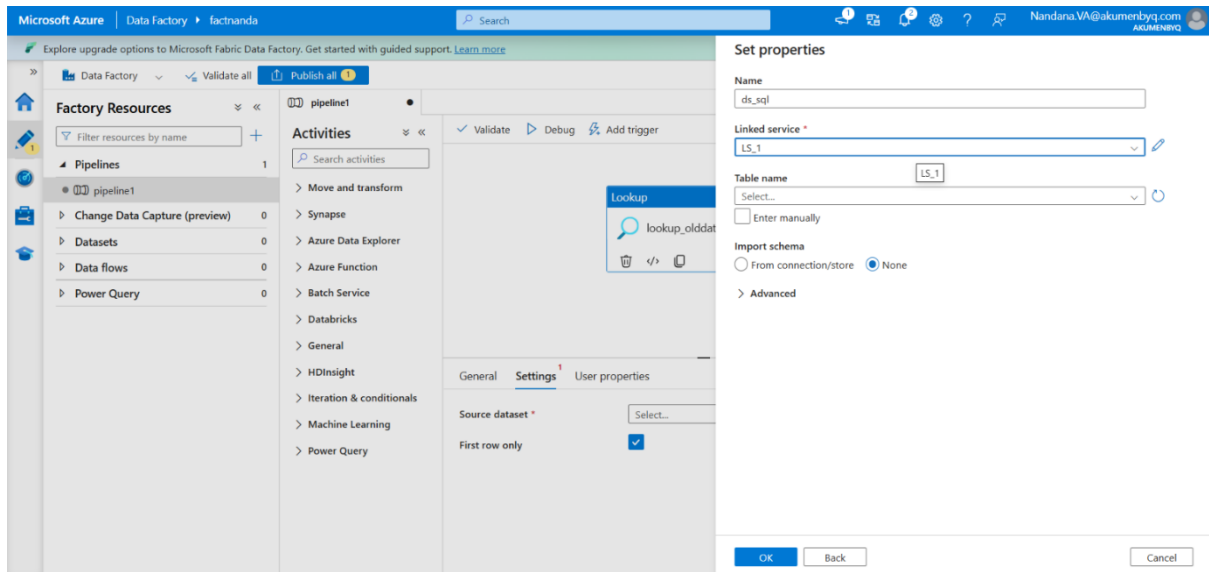
Verify the SQL Database and ADLS Linked Services.



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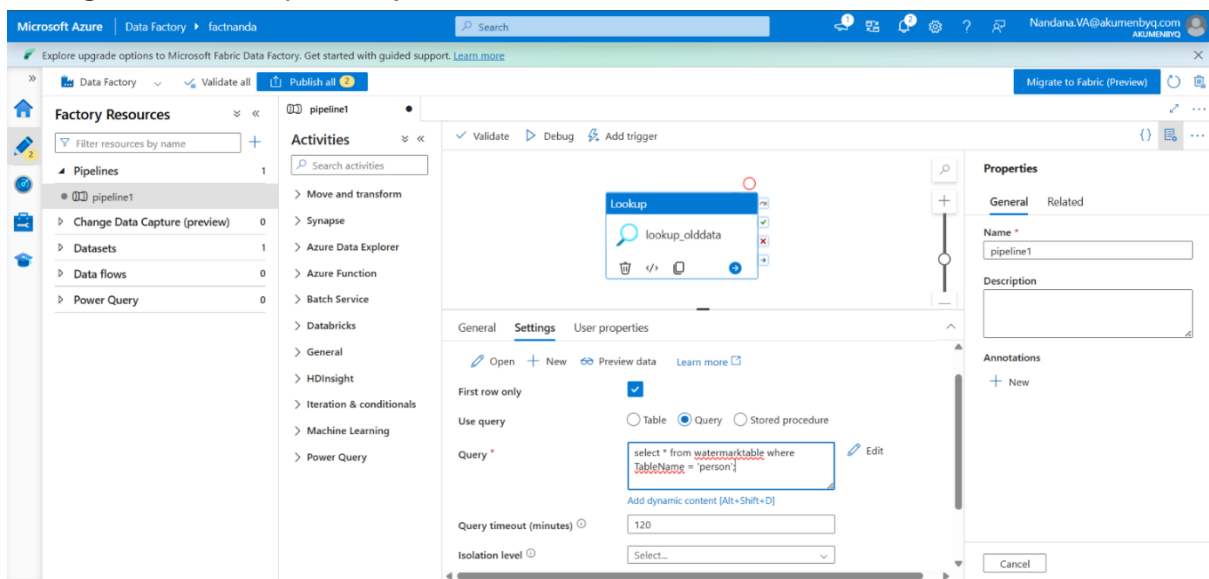
Configure the **Lookup activity** with the SQL Linked Service.

Select the required dataset/table.



Add the query to retrieve the **old watermark value**.

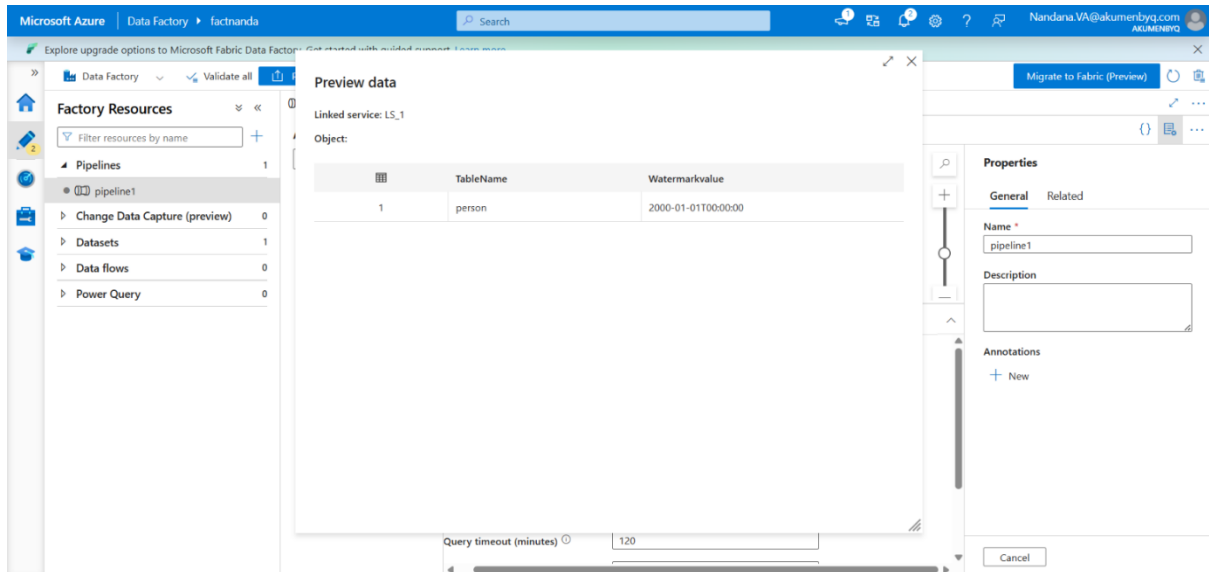
Configure the Lookup activity.



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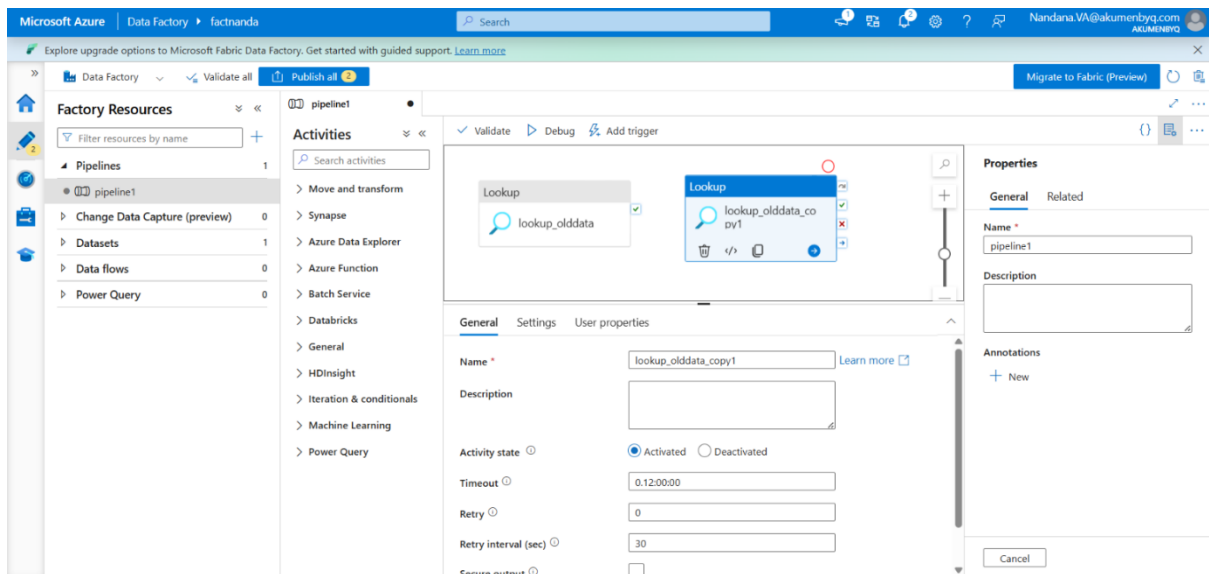
Click **Preview Data**.

Verify the old watermark value returned by the query.



Configure the second **Lookup** activity.

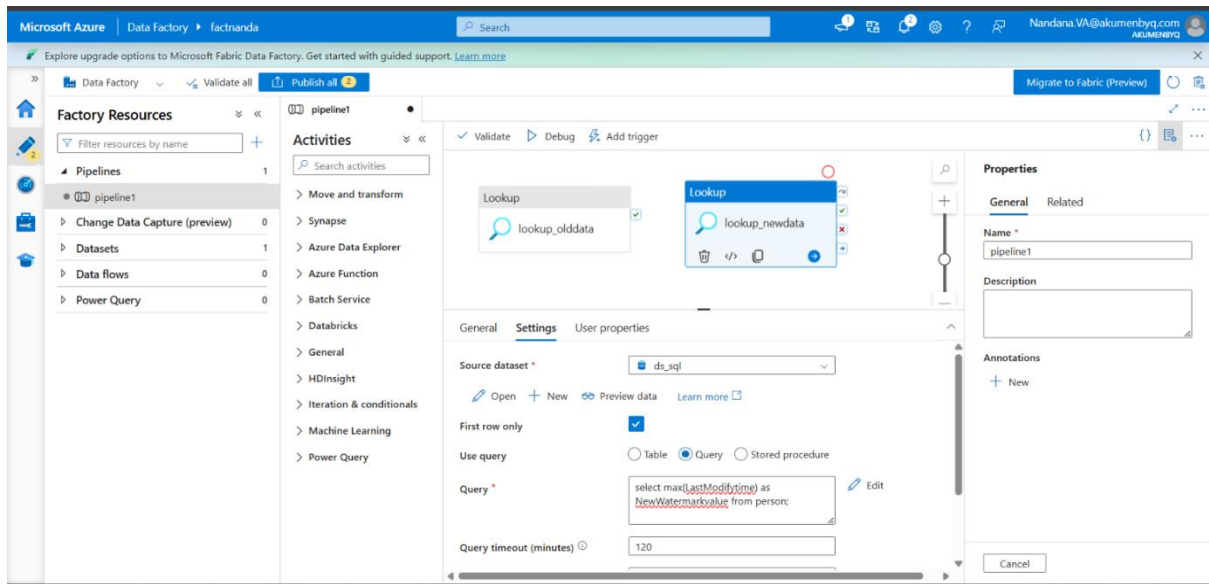
Use it to retrieve the new watermark value.



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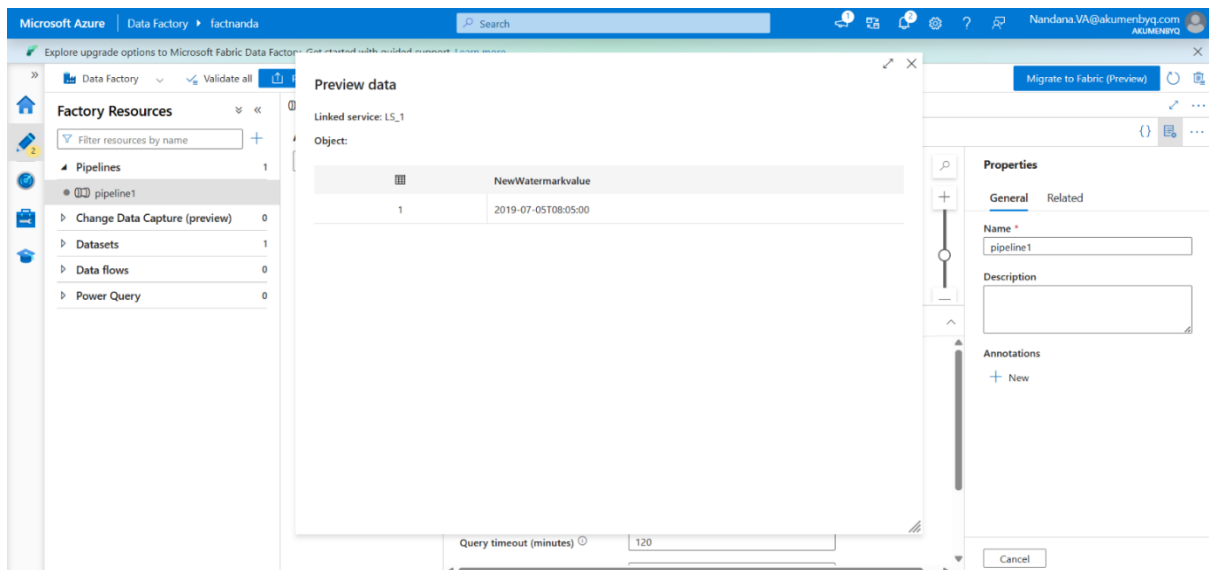
Configure the query to get the latest LastModifyTime.

Use it as the new watermark.



Click **Preview Data**.

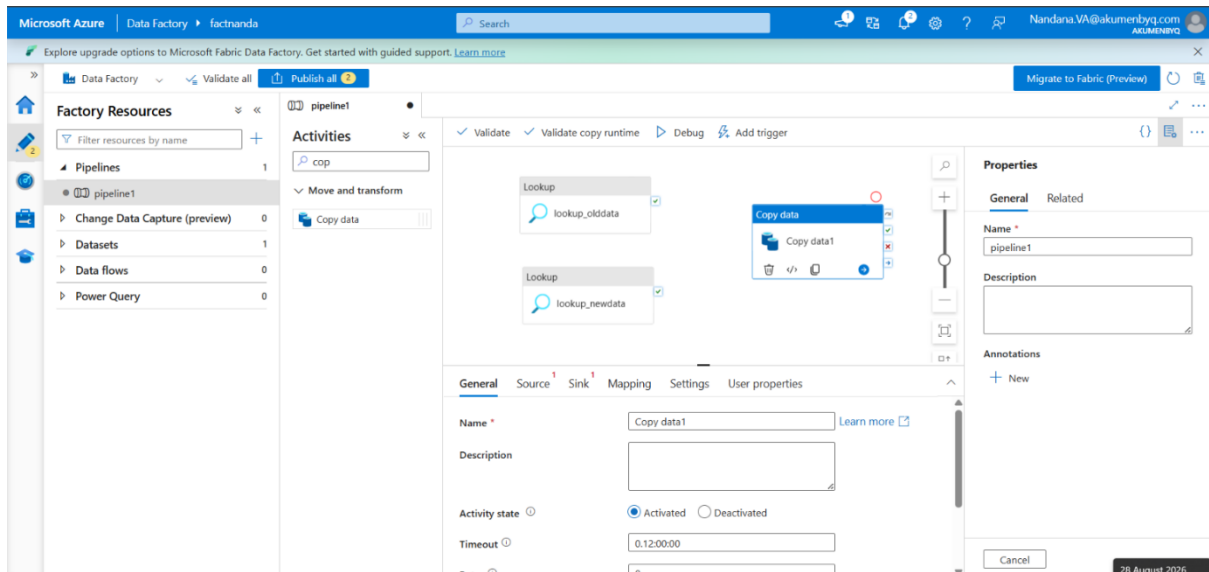
Verify the NewWatermarkValue.



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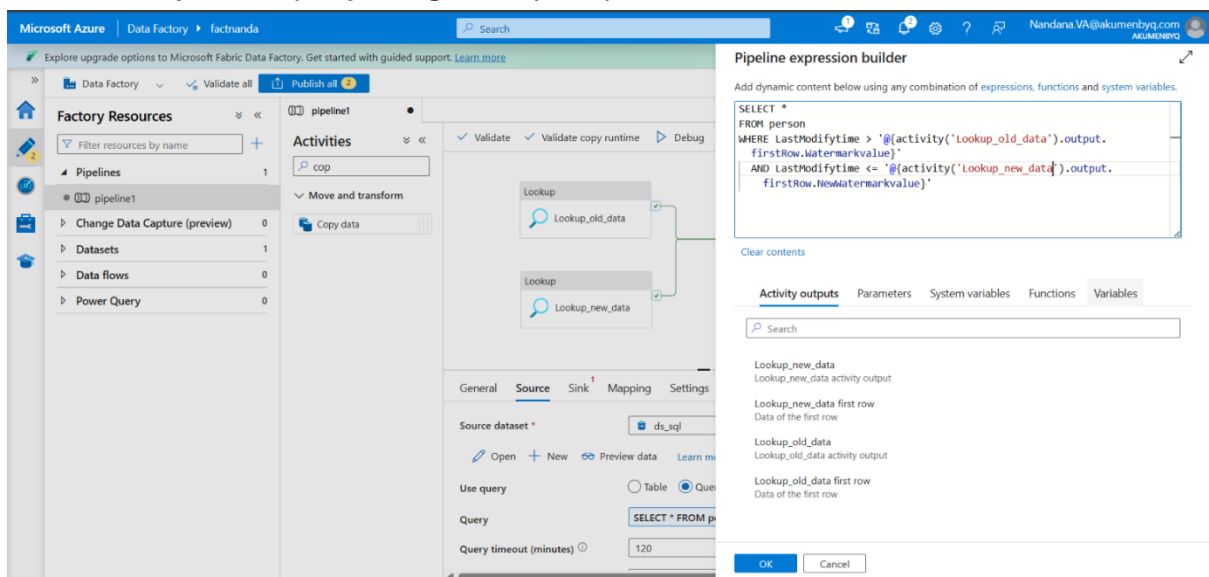
Add a **Copy Data** activity to the pipeline.

Connect it after the Lookup activities.



Open **Dynamic Content / Expression Builder**.

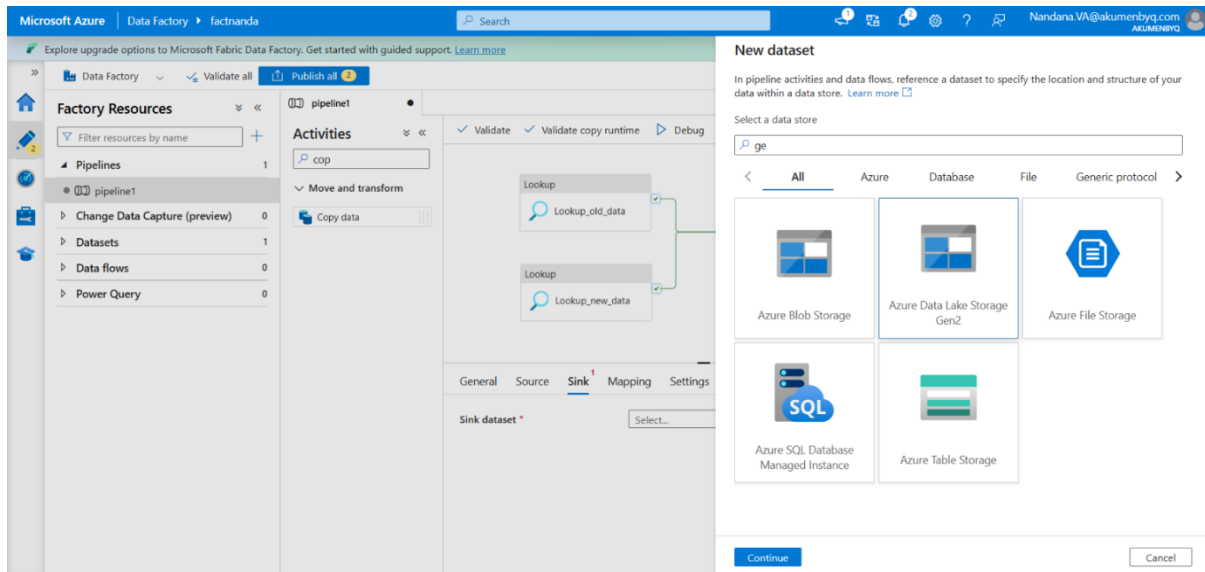
Create the dynamic query using Lookup output.



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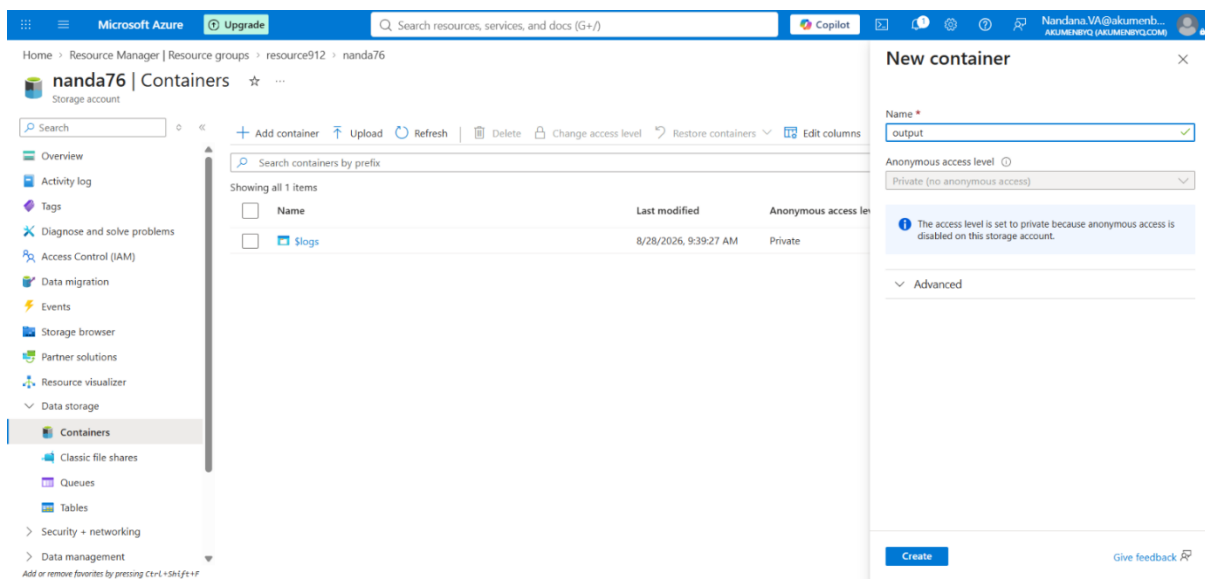
Create a new **Dataset** for ADLS Gen2.

Select the required Linked Service.



Create the required **ADLS Gen2 container**.

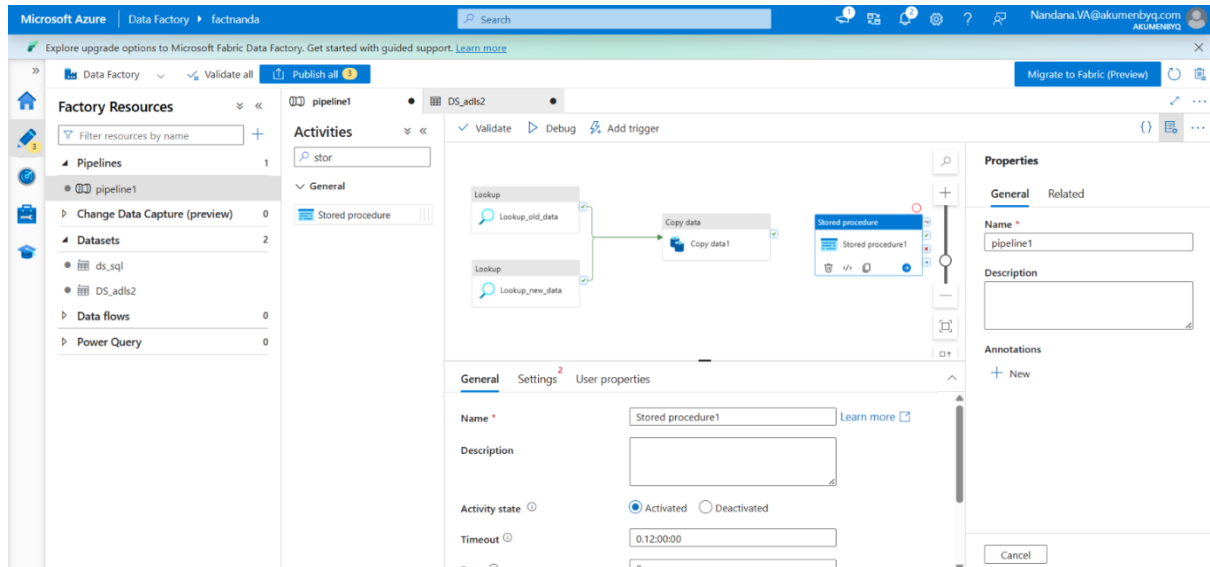
Use it as the destination for the incremental data.



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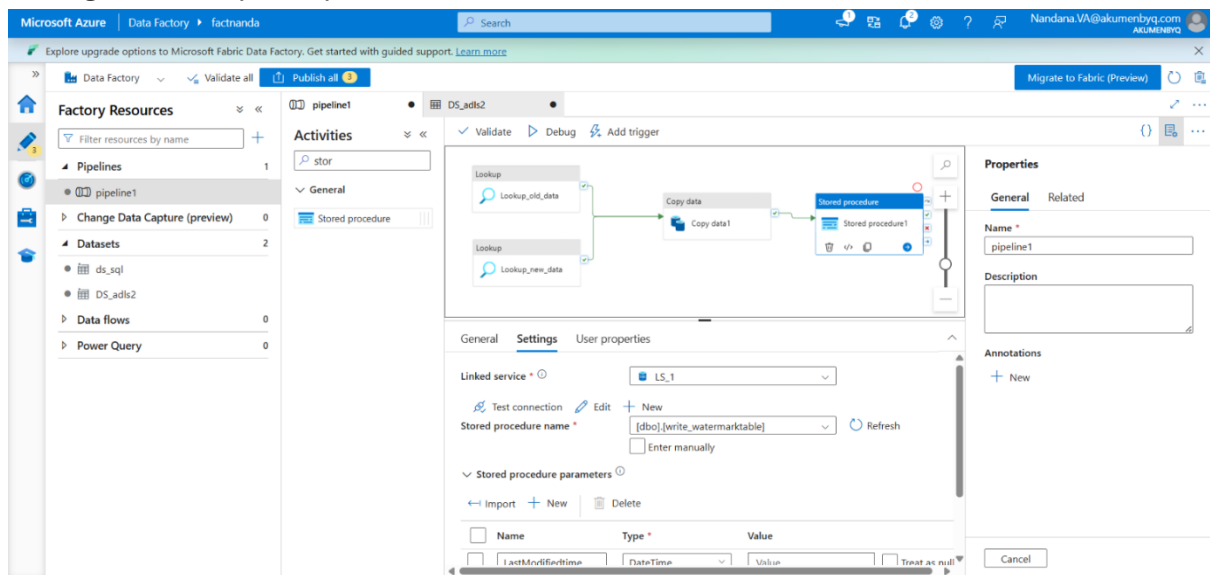
Add a **Stored Procedure activity** to the pipeline.

Connect it with the previous activity.



Select the Stored Procedure and Linked Service.

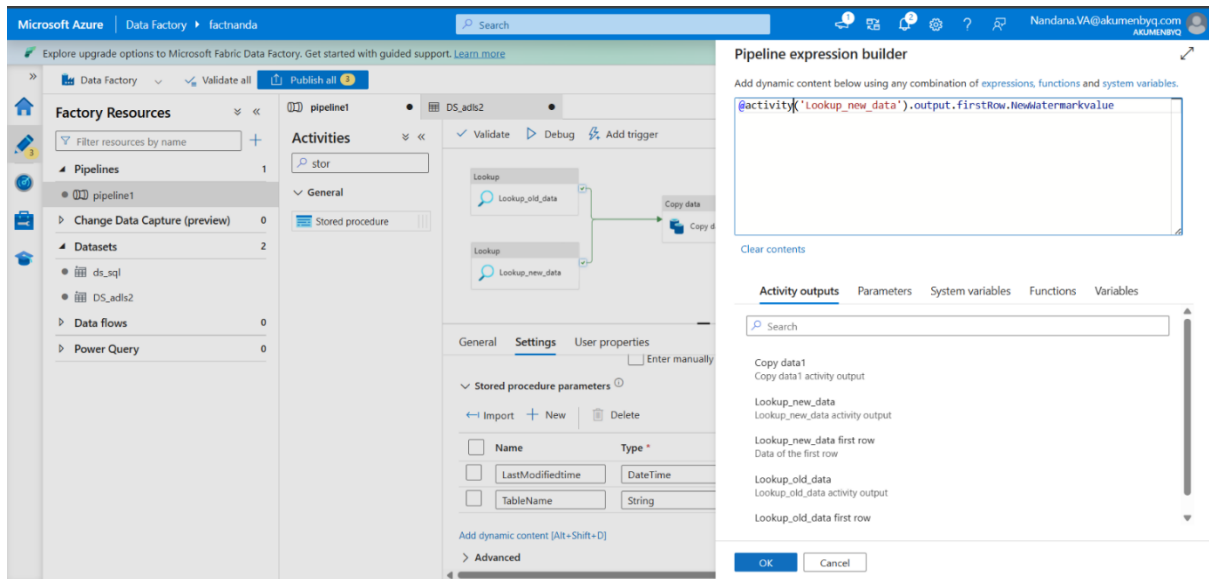
Configure the required parameters.



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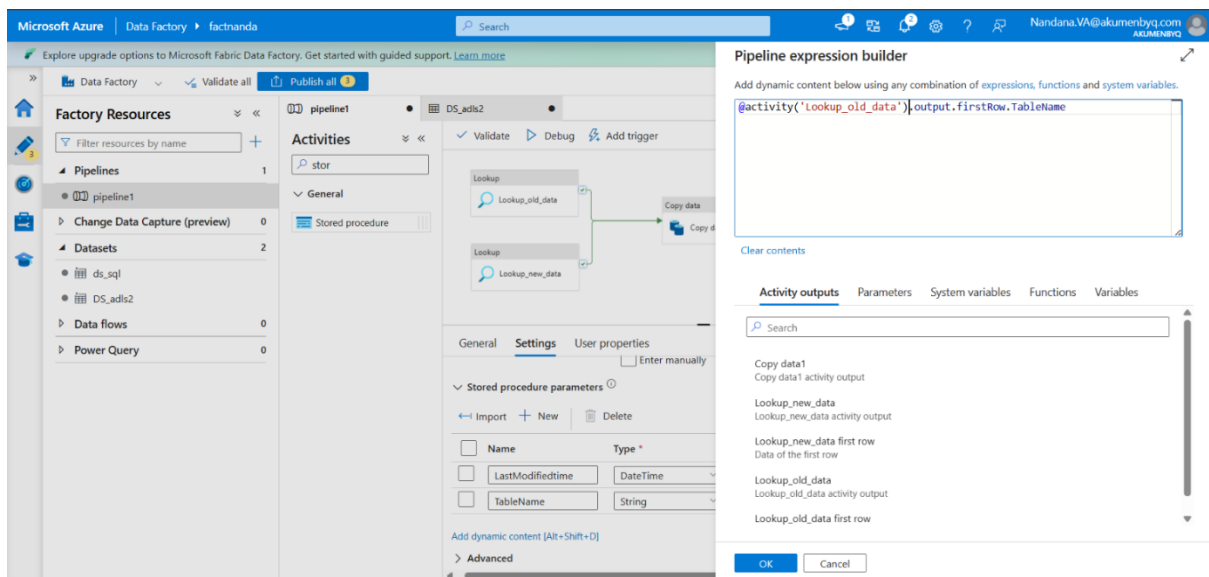
Open **Dynamic Content** for the Stored Procedure parameter.

Pass the new watermark value from the Lookup activity.



Pass the required **TableName** from the Lookup output.

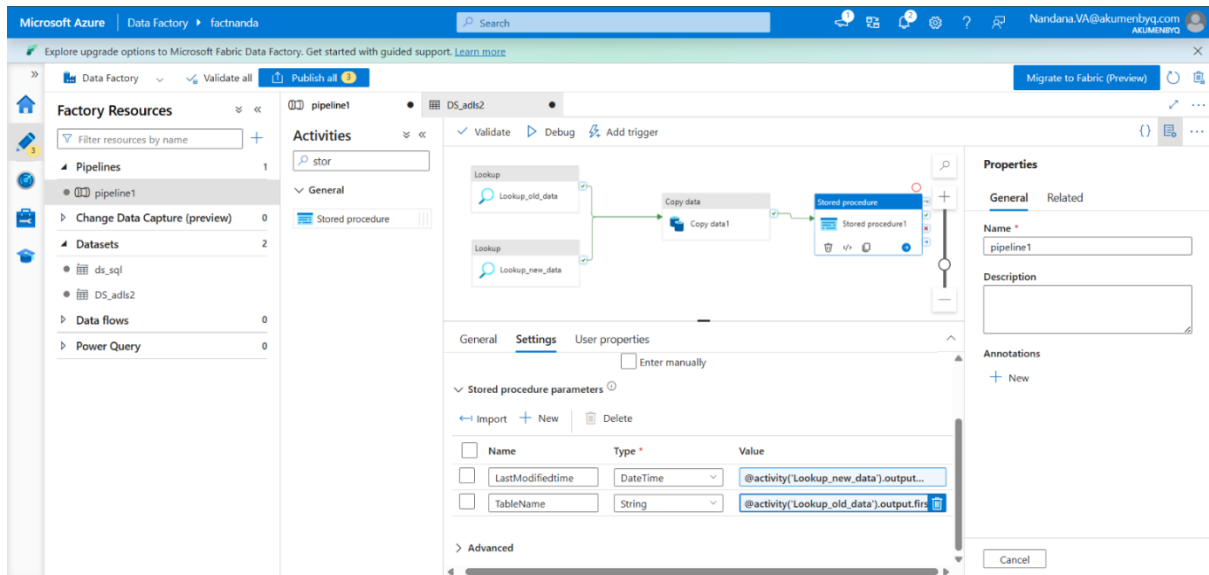
Use dynamic content.



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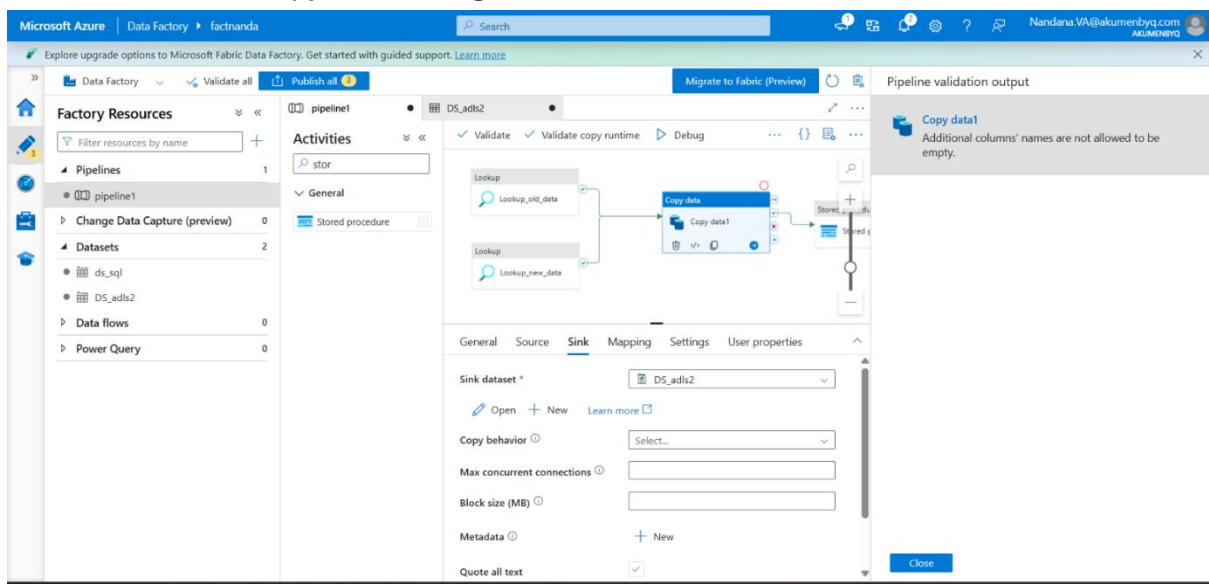
Verify all **Stored Procedure** parameters.

Check that the Lookup outputs are mapped correctly.



Validate the pipeline.

Check and fix the Copy Data configuration error.



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Run **Validate** again.

Confirm that the pipeline has no validation errors.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1), 'Datasets' (2), 'Data flows' (0), and 'Power Query' (0). The 'Pipelines' section is expanded, showing 'pipeline1'. The 'Activities' pane shows 'Lookup_old_data', 'Copy data', and 'Stored procedure'. The 'Stored procedure' activity is selected, and its settings are shown in the 'Settings' tab. The 'Linked service' is 'LS_1'. The 'Stored procedure name' is '[dbo].[write_watermarktable]'. The 'Stored procedure parameters' section is empty. On the right, the 'Pipeline validation output' pane displays a green checkmark and the message: 'Your pipeline has been validated. No errors were found.'

Click **Debug** to run the pipeline.

Open **Output** to check the execution result.

The screenshot shows the Microsoft Azure Data Factory interface with the pipeline 'pipeline1' running. The 'Activities' pane shows 'Lookup_old_data', 'Copy data', and 'Stored procedure'. The 'Stored procedure' activity is selected, and its settings are shown in the 'Settings' tab. The 'Linked service' is 'LS_1'. The 'Stored procedure name' is '[dbo].[write_watermarktable]'. The 'Stored procedure parameters' section is empty. On the right, the 'Output' pane displays the execution results. The output is a JSON object:

```
{  "firstRow": {    "NewWatermarkvalue": "2019-07-05T08:05:00Z"  },  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Southeast Asia)",  "billingReference": {    "activityType": "PipelineActivity",    "billableDuration": [  ]  ]}
```

 Below the JSON, a table shows the execution details for the 'Lookup' activity. The table has columns: 'Run start', 'Duration', and 'Integration runtime'. The table contains three rows of data.

Run start	Duration	Integration runtime
8/28/2026, 12:04:50 PM	26s	
8/28/2026, 12:04:27 PM	20s	AutoResolveIntegrationRuntime (East US)
8/28/2026, 12:04:04 PM	13s	AutoResolveIntegrationRuntime (Southeast Asia)

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Open the **Lookup activity Output**.

Verify the old watermark value returned.

The screenshot shows the Microsoft Azure Data Factory interface. The left sidebar displays the 'Factory Resources' tree with 'Pipelines' expanded, showing 'pipeline1'. The main canvas shows the 'Activities' section with 'Lookup' selected. The 'Output' window is open, displaying the output of the 'Lookup' activity. The output is a JSON object with the following structure:

```
{  "firstRow": {    "TableName": "person",    "Watermarkvalue": "2000-01-01T00:00:00Z"  },  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Southeast Asia)",  "billingReference": {    "activityType": "PipelineActivity"  }}
```

The 'Run start' and 'Duration' columns are visible in the table below the output window.

Open the **Copy Data activity Output**.

Check the data read and data written details.

The screenshot shows the Microsoft Azure Data Factory interface. The left sidebar displays the 'Factory Resources' tree with 'Pipelines' expanded, showing 'pipeline1'. The main canvas shows the 'Activities' section with 'Copy data' selected. The 'Output' window is open, displaying the output of the 'Copy data' activity. The output is a JSON object with the following structure:

```
{  "dataRead": 100,  "dataWritten": 220,  "filesWritten": 1,  "sourcePeakConnections": 1,  "sinkPeakConnections": 1,  "rowsRead": 5,  "rowsCopied": 5,  "copyDuration": 14,
```

The 'Run start' and 'Duration' columns are visible in the table below the output window.

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Open the **Stored Procedure activity Output**.

Verify the execution result.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1), 'Datasets' (2), 'Data flows' (0), and 'Power Query' (0). The 'Pipelines' section is expanded, showing 'pipeline1'. The 'Activities' pane shows 'Lookup' and 'Copy data'. The 'Output' pane displays the execution details for the 'Stored procedure' activity, including a table with columns: Activity name, Activity status, Activity type, Run start, Duration, and Integration runtime. The table shows four activities: 'Stored procedure1', 'Copy data1', 'Lookup_old_data', and 'Lookup_new_data', all of which are 'Succeeded'.

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
Stored procedure1	Succeeded	Stored procedu	8/28/2026, 12:04:50 PM	25s	AutoResolveIntegrationRuntime (Southe
Copy data1	Succeeded	Copy data	8/28/2026, 12:04:27 PM	20s	AutoResolveIntegrationRuntime (East US
Lookup_old_data	Succeeded	Lookup	8/28/2026, 12:04:04 PM	13s	AutoResolveIntegrationRuntime (Southe
Lookup_new_data	Succeeded	Lookup	8/28/2026, 12:04:04 PM	22s	AutoResolveIntegrationRuntime (Southe

Open the pipeline **Output**.

Check the execution status of all activities.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1), 'Datasets' (2), 'Data flows' (0), and 'Power Query' (0). The 'Pipelines' section is expanded, showing 'pipeline1'. The 'Activities' pane shows 'Lookup' and 'Copy data'. The 'Output' pane displays the execution details for the 'Stored procedure' activity, including a table with columns: Activity name, Activity status, Activity type, Run start, Duration, and Integration runtime. The table shows four activities: 'Stored procedure1', 'Copy data1', 'Lookup_old_data', and 'Lookup_new_data', all of which are 'Succeeded'.

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
Stored procedure1	Succeeded	Stored procedu	8/28/2026, 12:04:50 PM	25s	AutoResolveIntegrationRuntime (Southe
Copy data1	Succeeded	Copy data	8/28/2026, 12:04:27 PM	20s	AutoResolveIntegrationRuntime (East US
Lookup_old_data	Succeeded	Lookup	8/28/2026, 12:04:04 PM	13s	AutoResolveIntegrationRuntime (Southe
Lookup_new_data	Succeeded	Lookup	8/28/2026, 12:04:04 PM	22s	AutoResolveIntegrationRuntime (Southe

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Open **Copy Data** → **Details**.

Verify the source and destination data transfer.

The screenshot shows the 'Details' view of a Copy Data activity in Microsoft Azure Data Factory. The activity is named 'Copy data' and is located in the 'pipeline1' pipeline. The source is 'Azure SQL Database' (Region: Southeast Asia) and the destination is 'Azure Data Lake Storage Gen2' (Region: East US). The activity status is 'Succeeded'.

Activity run id: 2922cc2d-c532-487a-837f-bddc36be6a72

Data read: 100 bytes
Rows read: 5
Peak connections: 1

Data written: 220 bytes
Files written: 1
Rows written: 5
Peak connections: 1

Copy duration: 00:00:14
Throughput: 14 bytes/s

Start time: 8/28/2026, 12:04:28 PM
Used DUs: 4
Used parallel copies: 1
Duration: 00:00:14

Details:

	Working duration	Total duration
Queue	00:00:00	00:00:05
Transfer	00:00:00	00:00:07

Check the pipeline run status.

Review the displayed execution result.

The screenshot shows the 'Pipeline1' view in Microsoft Azure Data Factory. The pipeline is named 'pipeline1' and is located in the 'D5_adts2' dataset. The pipeline status is 'Canceled'.

Activities: Lookup, Copy data, Stored procedure

Output:

```
{
  "firstRow": {
    "TableName": "person",
    "Watermarkvalue": "2019-07-05T08:05:00Z"
  },
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Southeast Asia)",
  "billingReference": {
    "activityType": "PipelineActivity",
  }
}
```

pipeline status: Canceled

Run start	Duration	Integration runtime
8/28/2026, 12:09:46 PM	18s	AutoResolveIntegrationRuntime (East US)
8/28/2026, 12:09:30 PM	15s	AutoResolveIntegrationRuntime (Southeast Asia)
8/28/2026, 12:09:30 PM	12s	AutoResolveIntegrationRuntime (Southeast Asia)

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Run the SQL INSERT query.

Verify that the query executes successfully.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an Upgrade button, a search bar, and user information for Nandana.VA@akumenb... The main content area is titled 'sqlrdf (admin9/sqlrdf) | Query editor (preview)'. The left sidebar contains a navigation menu with options like Overview, Activity log, Tags, Diagnose and solve problems, Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings, Data management, Integrations, Power Platform, Security, Intelligent performance, Monitoring, Automation, and Help. The central pane shows the 'SQL query 4' editor with the following SQL code:

```
16
17 insert into person
18 values
19 (6, 'newrecord', '7/6/2019 9:05:00 AM'),
20 (7, 'newrecord', '7/6/2019 10:50:00 AM');
21
22 select * from person;
```

The Messages pane at the bottom shows the following message:

```
12:15:20 PM      Started executing on line 17
                  Affected rows: 2
                  Query executed successfully
```

A status bar at the bottom indicates 'Succeeded (0 sec 495 ms)'.

Run the SELECT query on the **Person** table.

Verify the records and LastModifyTime values.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an Upgrade button, a search bar, and user information for Nandana.VA@akumenb... The main content area is titled 'sqlrdf (admin9/sqlrdf) | Query editor (preview)'. The left sidebar contains a navigation menu with options like Overview, Activity log, Tags, Diagnose and solve problems, Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings, Data management, Integrations, Power Platform, Security, Intelligent performance, Monitoring, Automation, and Help. The central pane shows the 'SQL query 4' editor with the following SQL code:

```
17 insert into person
18 values
19 (6, 'newrecord', '7/6/2019 9:05:00 AM'),
20 (7, 'newrecord', '7/6/2019 10:50:00 AM');
21
22 select * from person;
```

The Results pane at the bottom shows the output of the SELECT query, displaying a table with columns PersonID, Name, and LastModifytime.

PersonID	Name	LastModifytime
2	Miya	2019-07-02T05:24:00.0000000
3	Ziya	2019-07-03T02:36:00.0000000
4	Niya	2019-07-04T03:20:00.0000000
5	Iiya	2019-07-05T08:05:00.0000000
6	newrecord	2019-07-06T09:05:00.0000000
7	newrecord	2019-07-06T10:50:00.0000000

A status bar at the bottom indicates 'Succeeded (0 sec 597 ms)'.

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Open the final pipeline **Output**.

Verify that the pipeline execution is successful.

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'pipeline1'. The main area shows the 'Activities' tab for 'pipeline1', displaying a workflow with 'Lookup' and 'Copy data' activities. The 'Output' window is open, showing the JSON output of the 'Copy data' activity, including 'firstRow' and 'effectiveIntegrationRuntime'. The 'Pipeline status' is 'Succeeded'. Below the status, a table lists the activities and their execution details:

Activity	Status	Run start	Duration	Integration runtime
Stored procedure1	Succeeded	8/28/2026, 12:17:39 PM	16s	AutoResolveIntegrationRuntime (Southeast Asia)
Copy data1	Succeeded	8/28/2026, 12:17:13 PM	24s	AutoResolveIntegrationRuntime (East US)
Lookup_new_data	Succeeded	8/28/2026, 12:16:43 PM	10s	AutoResolveIntegrationRuntime (Southeast Asia)
Lookup_old_data	Succeeded	8/28/2026, 12:16:43 PM	20s	AutoResolveIntegrationRuntime (Southeast Asia)

Open **Copy Data** → **Details**.

Verify the successful data transfer from SQL Database to ADLS Gen2.

The screenshot displays the 'Details' view of the 'Copy data' activity. It shows the activity run id: 336b554a-d019-46ca-8b01-611a69e2affa. The activity is a 'Copy' from 'Azure SQL Database' (Region: Southeast Asia) to 'Azure Data Lake Storage Gen2' (Region: East US). The status is 'Succeeded'. The details section shows the following metrics:

Metric	Value
Data read	60 bytes
Rows read	2
Peak connections	1
Data written	116 bytes
Files written	1
Rows written	2
Peak connections	1

The 'Duration' section shows the following details:

Category	Working duration	Total duration
Queue	00:00:00	00:00:09
Transfer	00:00:08	00:00:08